

Open X-Embodiment · arXiv 2310.08864 · Nov 2 · Data

Stakeholder

18+3

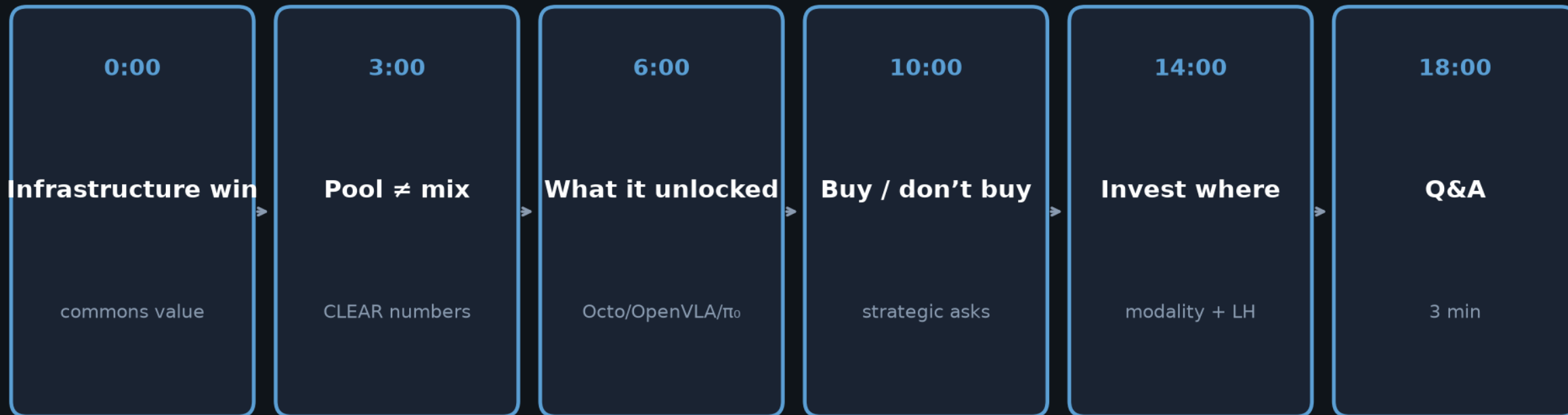
Infrastructure win — shared multi-lab pool
Cite RT-2-X as transfer proof, not production VLA

Paper club · Data cluster · due 9:30 AM PT

Spine: pool $\neq$ train mix · capacity $\times$ mix · steal mix science, don't drink the river

Open X-Embodiment Collaboration · 21 institutions · robotics-transformer-x.github.io

Stakeholder talk arc · 18+3



Spine: pool $\neq$ train mix · capacity $\times$ mix · steal mix science, don't drink the river

Stakeholder — infrastructure win

BUY / USE

- ✓ Standard RLDS download path
- ✓ Historical RT-1-X open baseline
- ✓ Syllabus: why every VLA cites OXE
- ✓ Eval harness mix for fair compares

DO NOT BUY

- × SOTA 2026 household (use $\pi_{0.5}$ /GR00T)
- × Guaranteed wrist / proprio / high-Hz
- × Balanced skill prior or hours metric
- × Production VLA from RT-2-X numbers

Strategic asks

1. Keep an OXE-derived mix in the eval harness — don't confound with private-only data.
2. Treat RT-2-X as transfer existence proof, not a production VLA.
3. Invest curation in modality coverage + long-horizon — not only more pick-place hours.
4. Cite as infrastructure win that enabled community generalists without each lab building PaLI-scale data.

CLEAR: Pool ≠ RT-X train mix

Marketing blur is the #1 citation failure mode

OXE POOL

Datasets	60
Embodiments	22
Institutions / labs	21 / 34
Trajectories	1M+
Skills / tasks	527 / 160,266
Scenes	311
Schema	RLDS TFRecords

RT-X TRAIN MIX

Manipulators	9 (≠ full 22)
Sources (ex.)	RT-1, QT-Opt, Bridge, Play, Jaco, Cable, ...
RT-1-X	robotics mix only
RT-2-X	~1:1 web : robotics
Eval	3600 trials / 6 robots
Downstream	Octo ~800k · OpenVLA ~970k

Moral: every claim needs a frozen mix card — pool size ≠ train size ≠ Octo/OpenVLA subset

Cite “22 robots / 1M+” for the commons; cite “9 manipulators” for RT-X experiment results

Headline numbers (tattoo)

Fig. 4

~+50%

relative mean SR lift
small-data domains
RT-1-X vs Original / RT-1

Table I

73% vs 92%

RT-1-X underfits
Google Robot 6 skills
capacity × mix

Table II

27.3 → 75.8

emergent skills ~3×
Bridge→Google Robot
X-mix transfer

Ablation

42.8%

no-Bridge mix
emergent collapses
Bridge is load-bearing

Table I — large native domains (exact)

Setting	Original	RT-1	RT-1-X	RT-2-X 55B
Bridge IRIS WidowX	13%	40%	27%	50%
Bridge RAIL WidowX	13%	30%	27%	30%
Google 6 skills	—	92%	73%	91%

RT-1-X hurts large in-domain SR — positive transfer is conditional on capacity

Fig. 4 moral — small-domain co-training lift

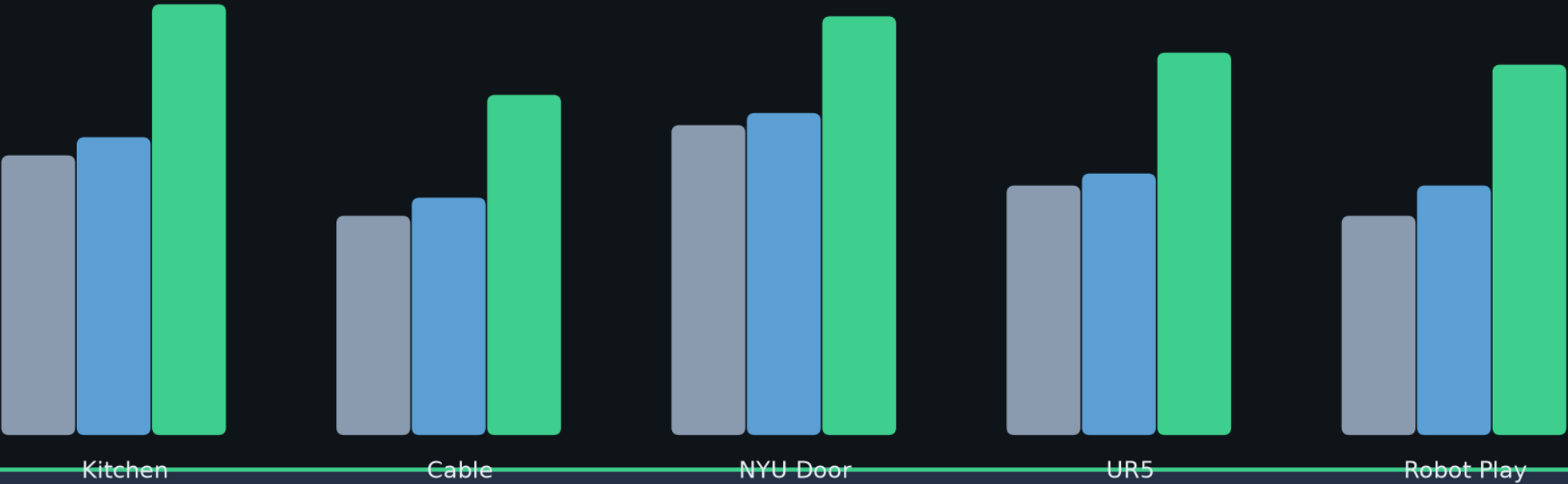
Matched architecture (RT-1 vs RT-1-X) isolates the mix as the variable

Legend

Original Method

RT-1 (single)

RT-1-X (mix)



CLEAR headline: RT-1-X mean success ~50% higher than Original or single-domain RT-1

Open X-Embodiment: Robotic Learning Datasets and RT-X Models

Open X-Embodiment Collaboration⁰

robotics-transformer-x.github.io

Abby O'Neill³⁴, Abdul Rehman³⁷, Abhinav Gupta⁴, Abhiram Maddukuri⁴⁵, Abhishek Gupta⁴⁶, Abhishek Padalkar¹⁰, Abraham Lee³⁴, Acorn Pooley¹¹, Agrim Gupta²⁸, Ajay Mandlekar²², Ajinkya Jain¹⁵, Albert Tong²⁸, Alex Bewley¹¹, Alex Herzog¹¹, Alex Irpan¹¹, Alexander Khazatsky²⁸, Anant Rai²³, Anchit Gupta¹⁹, Andrew Wang³⁴, Andrey Kolobov²⁰, Anikait Singh^{11,34}, Animesh Garg⁹, Aniruddha Kembhavi¹, Annie Xie²⁸, Anthony Brohan¹¹, Antonin Raffin¹⁰, Archit Sharma²⁸, Arefeh Yavary³⁵, Arhan Jain⁴⁶, Ashwin Balakrishna³², Ayzaan Wahid¹¹, Ben Burgess-Limerick²⁵, Beomjoon Kim¹⁷, Bernhard Schölkopf¹⁸, Blake Wulfe³², Brian Ichter¹¹, Cewu Lu^{27,8}, Charles Xu³⁴, Charlotte Le³⁴, Chelsea Finn^{11,28}, Chen Wang²⁸, Chenfeng Xu³⁴, Cheng Chi^{5,28}, Chenguang Huang³⁸, Christine Chan¹¹, Christopher Agia²⁸, Chuer Pan²⁸, Chuyuan Fu¹¹, Coline Devin¹¹, Danfei Xu⁹, Daniel Morton²⁸, Danny Driess¹¹, Daphne Chen⁴⁶, Deepak Pathak⁴, Dhruv Shah³⁴, Dieter Buehler¹⁸, Dinesh Jayaraman⁴², Dmitry Kalashnikov¹¹, Dorsa Sadigh¹¹, Edward Johns¹⁴, Ethan Foster²⁸, Fangchen Liu³⁴, Federico Ceola¹⁶, Fei Xia¹¹, Feiyu Zhao¹³, Felipe Vieira Fruejri²⁰, Freek Stulp¹⁰, Gaoyue Zhou²³, Gaurav S. Sukhatme⁴³, Gautam Salhotra^{43,15}, Ge Yan³⁶, Gilbert Feng³⁴, Giulio Schiavi⁷, Glen Berseth^{41,21}, Gregory Kahn³⁴, Guangwen Yang³³, Guanzhi Wang^{3,22}, Hao Su³⁶, Hao-Shu Fang²⁷, Haochen Shi²⁸, Henghui Bao⁴³, Heni Ben Amor², Henrik I Christensen³⁶, Hiroki Furuta³¹, Homanga Bharadhwaj^{4,19}, Homer Walke³⁴, Hongjie Fang²⁷, Huy Ha^{5,28}, Igor Mordatch¹¹, Ilija Radosavovic³⁴, Isabel Leal¹¹, Jacky Liang¹¹, Jad Abou-Chakra²⁵, Jaehyung Kim¹⁷, Jaimyn Drake³⁴, Jan Peters²⁹, Jan Schneider¹⁸, Jasmine Hsu¹¹, Jay Vakil¹⁹, Jeannette Bohg²⁸, Jeffrey Bingham¹¹, Jeffrey Wu³⁴, Jensen Gao²⁸, Jiaheng Hu³⁰, Jiajun Wu²⁸, Jialin Wu¹², Jiankai Sun²⁸, Jianlan Luo³⁴, Jiayuan Gu³⁶, Jie Tan¹¹, Jihoon Oh³¹, Jimmy Wu²⁴, Jingpei Lu³⁶, Jingyun Yang²⁸, Jitendra Malik³⁴, João Silvério¹⁰, Joey Hejna²⁸, Jonathan Bozher²⁸, Jonathan Tompson¹¹, Jonathan Yang²⁸, Jordi Salvador¹, Joseph J. Lim¹⁷, Junhyek Han¹⁷, Kaiyuan Wang³⁶, Kanishka Rao¹¹, Karl Pertsch^{34,28}, Karol Hausman¹¹, Keegan Go¹⁵, Keerthana Gopalakrishnan¹¹, Ken Goldberg³⁴, Kendra Byrne¹¹, Kenneth Oslund¹¹, Kento Kawaharazuka³¹, Kevin Black³⁴, Kevin Lin²⁸, Kevin Zhang⁴, Kiana Ehsani¹, Kiran Lekkala⁴³, Kirsty Ellis⁴¹, Krishan Rana²⁵, Krishnan Srinivasan²⁸, Kuan Fang³⁴, Kunal Pratap Singh⁶, Kuo-Hao Zeng¹, Kyle Hatch³², Kyle Hsu²⁸, Laurent Itti⁴³, Lawrence Yunliang Chen³⁴, Lerrel Pinto²³, Li Fei-Fei²⁸, Liam Tan³⁴, Linxi "Jim" Fan²², Lionel Ott⁷, Lisa Lee¹¹, Luca Weihs¹, Magnum Chen¹³, Marion Lepert²⁸, Marius Memmel⁴⁶, Masayoshi Tomizuka³⁴, Masha Itkina³², Mateo Guaman Castro⁴⁶, Max Spero²⁸, Maximilian Du²⁸, Michael Ahn¹¹, Michael C. Yip³⁶, Mingtong Zhang³⁹, Mingyu Ding³⁴, Minh Ho¹⁷, Mohan Kumar Srirama⁴, Mohit Sharma⁴, ...

Fig. 2 — Embodiments, scenes, trajectories, skill long-tail

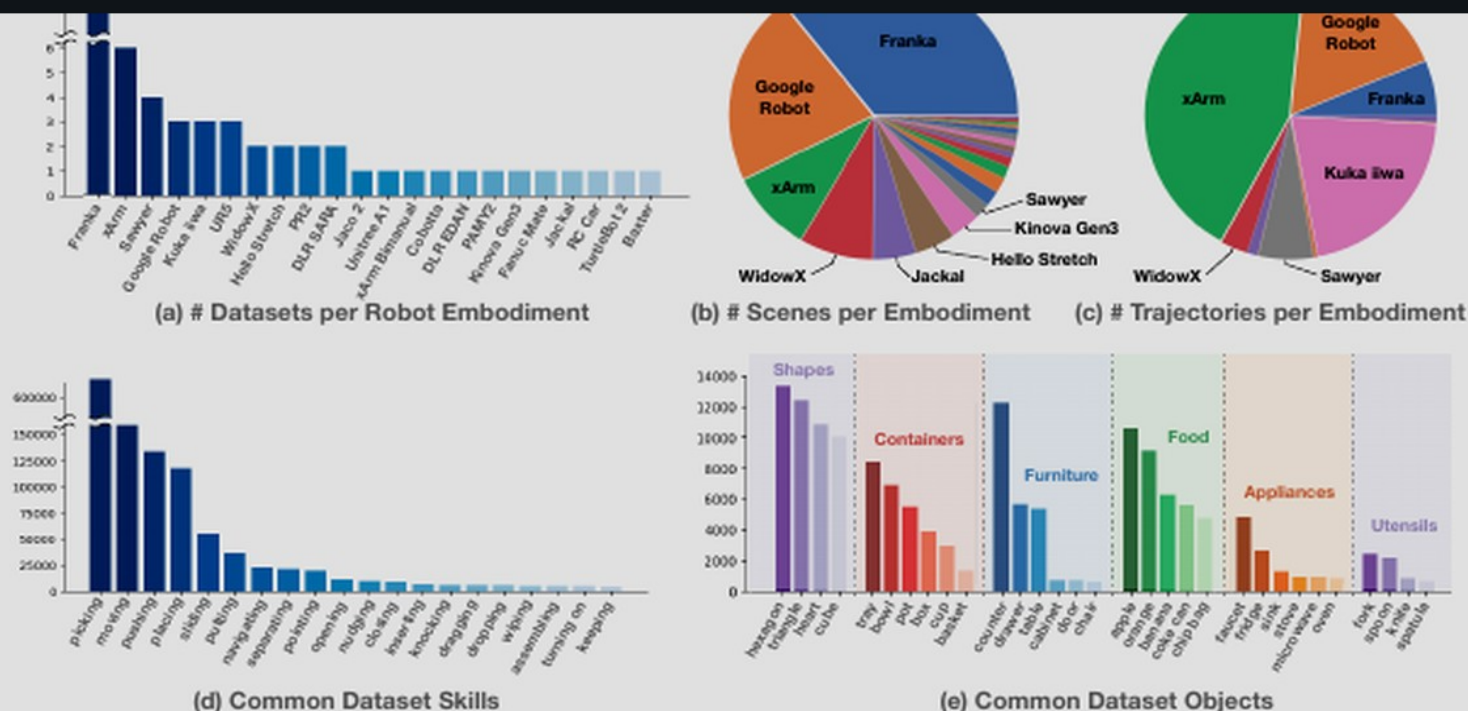


Fig. 2: The Open X-Embodiment Dataset. (a): the dataset consists of 60 individual datasets across 22 embodiments. (b): the Franka robot has the largest diversity in visually distinct scenes due to the large number of Franka datasets, (c): xArm and Google Robot contribute the most number of trajectories due to a few large datasets, (d, e): the dataset contains a great diversity of skills and common objects.

prior works, we directly train a policy on X-embodiment data, without any mechanisms to reduce the embodiment gap, and observe positive transfer by leveraging that data.

Large-scale robot learning datasets. The robot learning community has created open-source robot learning datasets, spanning grasping [60–71], pushing interactions [23, 72–74], sets of objects and models [75–85], and teleoperated demonstrations [8, 86–95]. With the exception of RoboNet [23], these datasets contain data of robots of the same type, whereas we focus on data spanning multiple embodiments.

- **Open X-Embodiment Dataset:** robot learning dataset with *1M+* robot trajectories from 22 robot embodiments.
- **Pre-Trained Checkpoints:** a selection of RT-X model checkpoints ready for inference and finetuning.

We intend for these resources to form a foundation for X-embodiment research in robot learning, but they are just the start. Open X-Embodiment is a community-driven effort, currently involving 21 institutions from around the world, and we hope to further broaden participation and grow the

Schema / consolidation used for RT-X

Observation

Image history + language; one canonical camera per dataset; unaligned poses

Action

7-DoF EE (x,y,z,r,p,y,grip) — abs/rel/vel as in source; NOT frame-aligned

Discretization

256 bins/dim · 8 outputs = 7 EE + episode terminate

Normalization

Per-dataset before binning; de-norm is embodiment-specific

Gap module

NONE — shared policy eats the soup; transfer is learned

RT-1-X

35M FiLM-EfficientNet + Transformer · robotics mix only · 3-10 Hz local

RT-2-X

PaLI-X VLA · actions as text tokens · ~1:1 web:robot · 5B/55B

Fig. 3 — RT-1-X / RT-2-X architecture + discrete EE

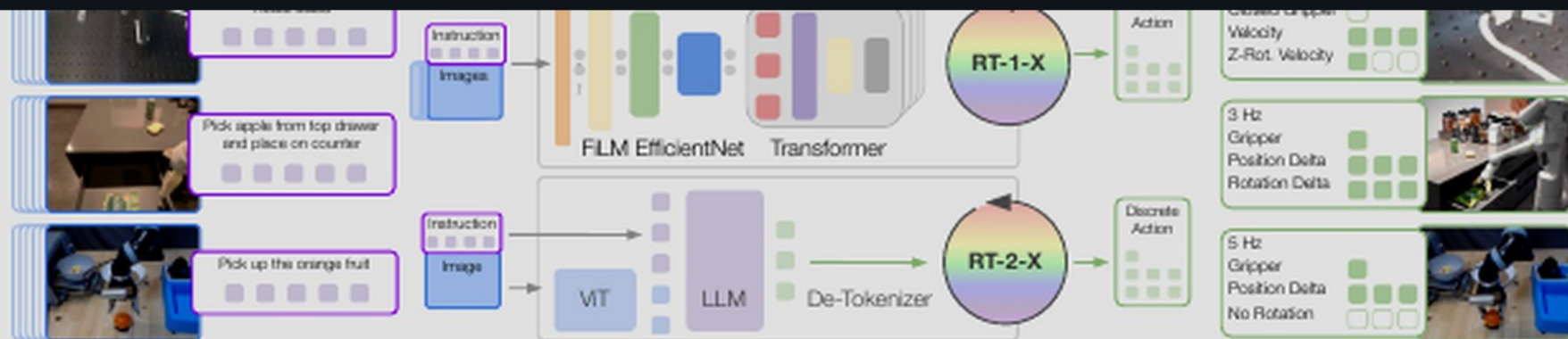


Fig. 3: RT-1-X and RT-2-X both take images and a text instruction as input and output discretized end-effector actions. RT-1-X is an architecture designed for robotics, with a FiLM [116] conditioned EfficientNet [117] and a Transformer [118]. RT-2-X builds on a VLM backbone by representing actions as another language, and training action text tokens together with vision-language data.

metadata) per embodiment (Fig. 2(b)), where Franka dominates. Fig. 2(c) shows the breakdown of trajectories per embodiment. To further analyze the diversity, we use the language annotations present in our data. We use the PaLM language model [3] to extract objects and behaviors from the instructions. Fig. 2(d,e) show the diversity of skills and objects. While most skills belong to the pick-place family, the long tail of the dataset contains skills like “wiping” or “assembling”. Additionally, the data covers a range of household objects, from appliances to food items and utensils.

IV. RT-X DESIGN

To evaluate how much X-embodiment training can improve the performance of learned policies on individual robots, we require models that have sufficient capacity to productively make use of such large and heterogeneous

B. Policy architectures

We consider two model architectures in our experiments: (1) RT-1 [8], an efficient Transformer-based architecture designed for robotic control, and (2) RT-2 [9] a large vision-language model co-fine-tuned to output robot actions as natural language tokens. Both models take in a visual input and natural language instruction describing the task, and output a tokenized action. For each model, the action is tokenized into 256 bins uniformly distributed along each of eight dimensions; one dimension for terminating the episode and seven dimensions for end-effector movement. Although both architectures are described in detail in their original papers [8, 9], we provide a short summary of each below:

RT-1 [8] is a 35M parameter network built on a Transformer architecture [118] and designed for robotic control,

Capacity × mix co-design (Table I moral)

More data is not free — small models underfit rich X-mixes on large native domains

RT-1-X (35M) on X-mix

Google 6-skill: 73% (vs RT-1 92%)

Bridge IRIS: 27% (vs RT-1 40%)

Underfits large native sets

RT-2-X (55B) on X-mix

Google 6-skill: 91% (recovers)

Bridge IRIS: 50% (beats)

Capacity absorbs diversity

BEHAVIOR implication

- If shipping $\pi_{0.5}$ / GROOT-scale → you can afford OXE-like width (with weight table).
- If shipping a tiny policy → curate narrower; gate external shards behind validation q-score.
- Empiricist underfit probe: same heavy mix, small vs base open ckpt, matched steps.

strong performance in these two evaluation scenarios.

Table [91] datasets. RT-1-X is trained on only robotics mixture data defined above, whereas RT-2-X is trained via co-fine-tuning (similarly to the original RT-2 [9]), with an approximately one to one split of the original VLM data and the robotics data mixture. Note that the robotics data mixture used in our experiments includes 9 embodiments which is fewer than the entire Open X-Embodiment dataset (22) – the practical reason for this difference is that we have continued to extend the dataset over time, and at the time of the experiments, the dataset above represented all of the data. In the future, we plan to continue training policies on the extended versions of the dataset as well as continue to grow the dataset together with the robot learning community.

At inference time, each model is run at the rate required for the robot (3-10 Hz), with RT-1 run locally and RT-2 hosted on a cloud service and queried over the network.

V. EXPERIMENTAL RESULTS

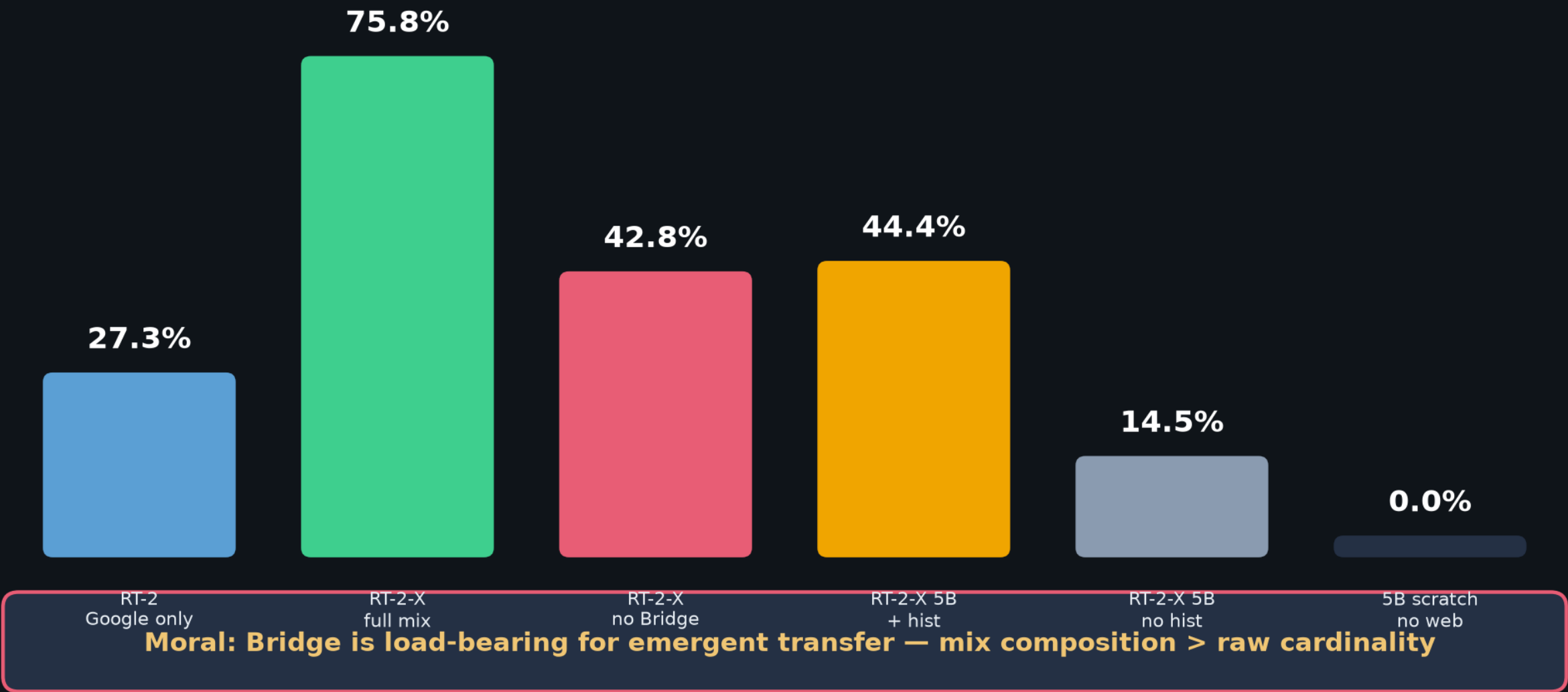
Our experiments answer three questions about the effect of X-embodiment training: (1) Can policies trained on our

tion and robot embodiment as in the respective publications. For large-scale dataset experiments, we consider Bridge [95] and RT-1 [8] for in-distribution evaluation and use their respective robots: WidowX and Google Robot.

For each small dataset domain, we compare the performance of the RT-1-X model, and for each large dataset we consider both the RT-1-X and RT-2-X models. For all experiments, the models are co-trained on the full X-embodiment dataset. Throughout this evaluation we compare with two baseline models: (1) The model developed by the creators of the dataset trained only on that respective dataset. This constitutes a reasonable baseline insofar as it can be expected that the model has been optimized to work well with the associated data; we refer to this baseline model as the *Original Method* model. (2) An RT-1 model trained on the dataset in isolation; this baseline allows us to assess whether the RT-X model architectures have enough capacity to represent policies for multiple different robot platforms simultaneously, and whether co-training on multi-embodiment data leads to higher performance.

Table II — Bridge ablation (emergent skills)

Bridge→Google Robot skills absent from Google-only data



Genealogy — OXE → soups → BEHAVIOR



What OXE replaced

Single-lab teleop as default pretrain
“One robot, one policy” as only recipe

What OXE did NOT replace

Post-train gold demos · wrist/proprio
Continuous chunks · challenge sim alignment

Children drink the water: Octo compares to RT-1-X · OpenVLA beats RT-2-X on open suite

π_0 open share $\approx$ 9.1% OXE Magic Soup + Bridge + DROID · BEHAVIOR needs OXE-scale thinking, not raw dump

Red lines + municipal water metaphor

OXE = MUNICIPAL WATER

Shared multi-lab supply
Enabled RT-X / Octo / OpenVLA / π_0
Agglomeration + RLDS schema
Positive transfer existence proof

BEHAVIOR'26 = BOTTLED BLEND

Specialty challenge mix
Stage-rich · wrist-rich · schema-clean
~20k gold demos as post-train depth
Steal science — don't drink the river

DO NOT

- × Dump raw OXE into BEHAVIOR FT at uniform weight
- × Assume discrete 7×256 bins are enough for 2026 long-horizon household
- × Claim “OXE alone = BEHAVIOR'26 data plan”
- × Cite unpublished OXE hours
- × Skip mix-card discipline / transfer ablations

Fig. 4 — Small-data domains: RT-1-X ~+50% mean lift

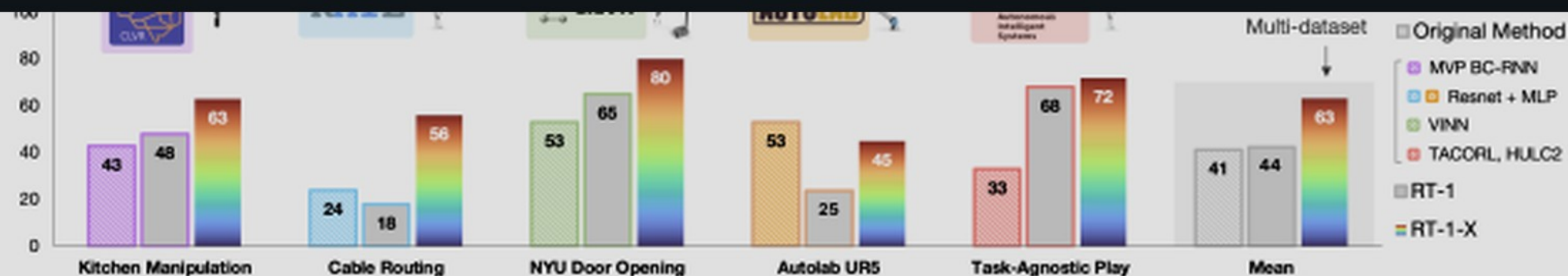


Fig. 4: RT-1-X mean success rate is 50% higher than that of either the Original Method or RT-1. RT-1 and RT-1-X have the same network architecture. Therefore the performance increase can be attributed to co-training on the robotics data mixture. The lab logos indicate the physical location of real robot evaluation, and the robot pictures indicate the embodiment used for the evaluation.

Evaluation Setting	Bridge	Bridge	RT-1 paper 6 skills
Evaluation Location	IRIS (Stanford)	RAIL Lab (UCB)	Google Robotic Lab
Robot Embodiment	WidowX	WidowX	Google Robot
Original Method	LCBC [95]	LCBC [95]	-
Original Method	13%	13%	-
RT-1	40%	30%	92%
RT-1-X	27%	27%	73%
RT-2-X (55B)	50%	30%	91%

TABLE I: Parameter count scaling experiment to assess the impact of capacity on absorbing large-scale diverse embodiment data. For these large-scale datasets (Bridge and RT-1 paper data), RT-1-X underfits and performs worse than the Original Method and RT-1. RT-2-X model with significantly many more parameters can obtain strong performance in these two evaluation scenarios.

tasks. We split our evaluation into two types: evaluation on domains that have small-scale datasets (Fig. 4), where we would expect transfer from larger datasets to significantly improve performance, and evaluation on domains that have large-scale datasets (Table I), where we expect further improvement to be more challenging. Note that we use the same robotics data *training* mixture (defined in Sec. IV-C) for all the evaluations presented in this section. For small-scale dataset experiments, we use Kitchen Manipulation [128], Cable Routing [129], NYU Door Opening [130], AUTOLab UR5 [132], and Robot Play [134]. We use the same evaluation and robot embodiment as in the respective publications. For large-scale dataset experiments, we consider Bridge [95]

Stakeholder · close

Infrastructure win — commons enabled open generalists **Invest modality + long-horizon, not only pick-place hours**

Cite as shared pool success · RT-2-X = existence proof
Keep OXE-derived mix in eval harness

Q&A · keep the spine clear

Open X-Embodiment

Robotic Learning Datasets and RT-X Models

arXiv 2310.08864 · Nov 2 Data cluster

OXE = municipal water · BEHAVIOR'26 = bottled specialty blend

Steal mix science — don't drink the river

Pool: 60 ds / 22 emb / 1M+ traj Train mix: 9 manipulators Eval: 3600 trials / 6 robots

Headlines: Fig.4 ~+50% · Table I RT-1-X 73% vs RT-1 92% · emergent 27.3%→75.8% · no-Bridge 42.8%

Seat 1 Visionary = Kanta Saito · Quiet build